S e n t r i q

Jurisdiction-Aware AML/KYC Transaction Monitoring

A prototype RegTech tool purpose-built for private wealth management
Entity: Julius Baer | US (FinCEN/BSA) vs Singapore (MAS Notice 626)

MH6822 Regulatory Technology — Assignment 1, Task 3

THE CORE PROBLEM

No incumbent TM platform handles regulatory divergence across jurisdictions automatically.

01 Fixed thresholds applied uniformly across all jurisdictions — producing 90 to 95% false positive rates in private banking TM

02 No native switching between regulatory regimes — every update requires an IT change request, leaving the bank exposed during the gap

03 Account-level screening cannot see layering across BVI holdcos, family trusts, and fund subscriptions linked to the same beneficial owner

What a Memo or Spreadsheet Cannot Do

WHAT PROBLEM DOES THIS SOLVE

A memo or spreadsheet can...	Sentriq can...
Record a threshold decision	Apply different PEP EDD logic depending on booking centre
List which rules apply per country	Detect layering across 5 linked client accounts
Document an alert closure	Score transactions at scale and explain every result
Summarise regulatory differences	**Auto-generate the STR narrative MAS requires for STRO filing**

Why Sentriq is Right for Julius Baer

QUESTION 1: WHY THIS PRODUCT

CHF 460B AUM	3 Booking Centres	DOJ History
Pure-play private bank HNWI-only client base BVI, trusts, foundations are standard structures	Singapore, Hong Kong New York, Geneva HQ Each under a different regulatory regime	2016 USD 547M settlement US regulatory exposure is real and documented not hypothetical

Off-the-shelf tools — NICE Actimize, Oracle FCCM, IMTF — apply fixed thresholds across all jurisdictions simultaneously. For Julius Baer, where a single client may hold accounts across a BVI holdco, a Jersey trust, and a Cayman fund, this produces two specific failures: 90 to 95% of alerts are false positives, and genuine layering across linked entities goes undetected entirely.

Two-Layer Scoring — Rules Plus Anomaly Detection

HOW IT WORKS

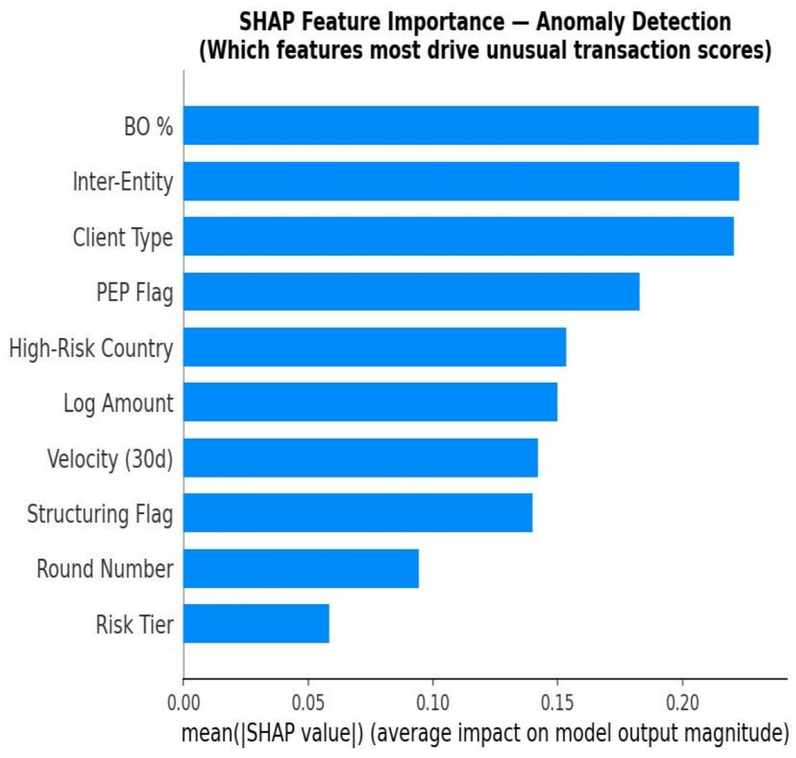
LAYER 1 — Rule-Based Engine (60%)

6 rules, each applying different logic per jurisdiction:
CTR threshold | Structuring | PEP (foreign vs domestic)
High-risk country | BO gap | Inter-entity transfer

LAYER 2 — Isolation Forest (40%)

Catches typologies no rule covers.
SHAP explains which features drove each score,
supporting the MAS STR narrative requirement.

OUTPUT: Score 0-100 | RED / AMBER / GREEN | Jurisdiction narrative



SHAP Feature Importance — top drivers of anomaly score

Three Key Differences — US vs Singapore

QUESTION 3: REGULATORY DIVERGENCE

1. Alert Basis

US

Threshold-driven
CTR mandatory above USD 10,000
Amount alone triggers filing obligation

SG

Risk-based
No fixed threshold
Suspicion — not amount — triggers STR

Why it matters:

Same USD 750,000 wire: US files a CTR automatically. SG analyst must assess risk and write a narrative.

2. PEP Scope

US

Foreign PEPs only
Domestic PEPs historically excluded from mandatory EDD under BSA

SG

Foreign AND domestic PEPs
Former Singapore civil servants and their families — EDD mandatory

Why it matters:

Same client (former SG civil servant) booked in New York: no EDD required. Booked in Singapore: EDD mandatory.

3. Beneficial Ownership

US

Flat 25% threshold
FinCEN CDD Rule — fixed regardless of client risk tier

SG

Risk-based threshold
Down to 10% for high-risk clients
Trusts and foundations scrutinised deeper

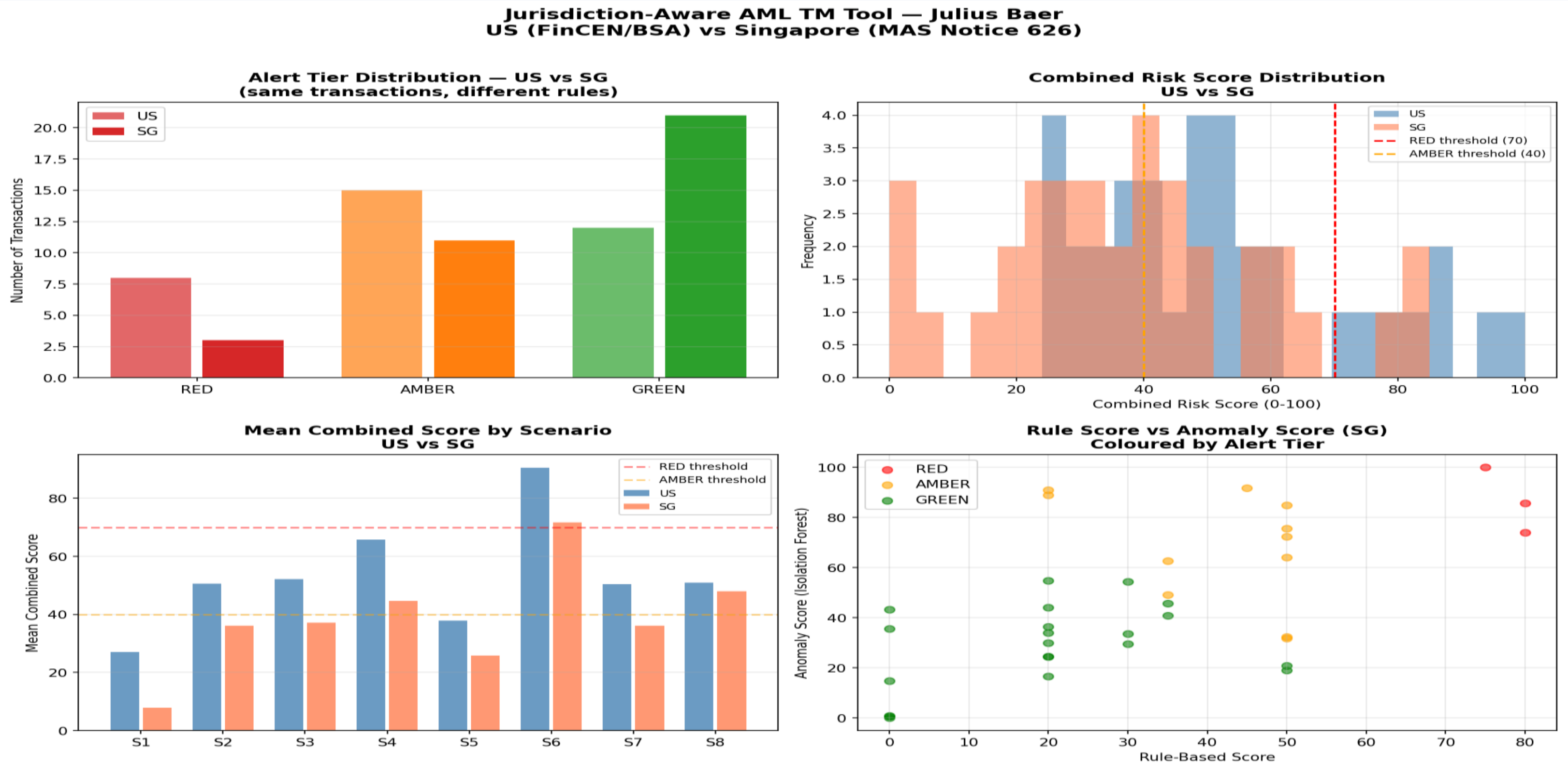
Why it matters:

BO disclosed at 15%: passes US FinCEN rule. Fails MAS Notice 626 for a high-risk trust or foundation.

The jurisdiction config switches all parameters simultaneously — no IT change request, no regulatory risk gap.

Jurisdiction Divergence in Practice

LIVE TOOL OUTPUT



US: 8 RED | 15 AMBER | 12 GREEN   SG: 3 RED | 11 AMBER | 21 GREEN

5 transactions RED in US only (threshold-driven) | 0 RED in SG only | 3 RED in both

Same Transaction. Two Completely Different Outputs.

JURISDICTION DIVERGENCE — LIVE EXAMPLE

TX-S4-003 | Petrov Capital Ltd (BVI) | Wire USD 750,000 | Scenario: BVI Holdco Layering

US | Score 83 | RED

Alert basis: Threshold
CTR required: USD 750K > USD 10K

Rules fired:
CTR threshold
BO breach (15% under 25%)
Inter-entity transfer

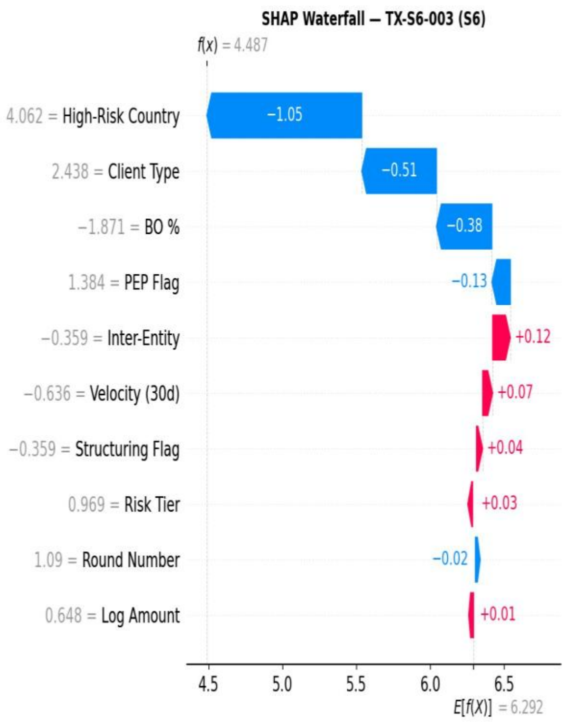
Action: SAR filing recommended.
Escalate to MLRO.

SG | Score 59 | AMBER

Alert basis: Risk-based

Auto-generated STRO narrative:
Flagged because: inter-entity transfer with no documented commercial rationale; elevated velocity (6 txns/30d). Reasonable grounds to suspect proceeds of unlawful activity.

Action: Further due diligence.
Document rationale.



SHAP waterfall: features driving TX-S6-003 anomaly score

Boundaries — And Why We Drew Them Here

QUESTION 4: WHAT IT DOES NOT DO

Does not replace MLRO judgement

Filing a SAR or STR is a legal act requiring human accountability. The tool triages and explains. The decision to file always remains with the compliance officer. This boundary is principled, not a technical gap.

Does not cover FINMA, HKMA, or EU AMLD6

Julius Baer's Geneva HQ and Hong Kong operations were acknowledged as the honest next milestones. Adding them requires additional regulatory research. Overstating coverage would be a worse failure than scoping honestly.

No live sanctions screening

OFAC SDN and UN Consolidated List integration requires real-time API access beyond prototype scope. In production this would be the first integration after validation.

Synthetic data and real-world data quality risks

Built on a small synthetic sample for prototype purposes. Real data performance is dependent on data quality from production systems — missing BO fields, inconsistent PEP flags, and incomplete entity linkage are common in practice. Production deployment requires clean, validated data, retraining on real transaction history, and periodic model revalidation.

The Case for Sentriq

1. Purpose-built for HNWI complexity — trusts, BVI holdcos, Cayman funds, multi-entity structures across booking centres
2. Jurisdiction-aware by architecture — switching US to SG changes six regulatory parameters simultaneously, no IT ticket required
3. Two-layer scoring catches both known typologies and statistically anomalous patterns rules alone cannot detect
4. SHAP explainability directly supports the MAS Notice 626 STR narrative requirement — analyst does not start from a blank page
5. Honest about boundaries — human judgement is never outsourced, scope is declared not overstated